Forest Inventory

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Forest Inventory (A11622)

Short Title:	Forest Inventory		
Faculty	Science and Computing		
Department	Science		
Cluster	Forestry		
Author	MPEDINI		
Credits	10	**Level:**	Intermediate

Description of Module / Aims

In this (10 credit) module students will carry out a wide range of forest inventory work in a medium-sized forest property over a number of days. Students will plan and carry out field work in teams. Social, environmental and economical values will be mapped in a Geographic Information System (GIS), and a suite of maps and an Environmental Risk Assessment for the property will be prepared by students. Sampling theory and statistical methods will be introduced in class, followed by stand volume estimation field work in the forests. At the end of the semester students will have written three short forest inventory-related reports to a professional standard, which will constitute the foundation of a forest management plan.

Programmes

`FORS-0011`	BSc in Forestry (`WD_SFORS_D`)	3 / 5 / M

Indicative Content

- Forest inventory procedures relating to social, environmental and economic values
- Stand mapping best practice
- Collection of data for Environmental Risk Assessment
- Geodatabase design for forest inventory and management planning
- Map design and layout including advanced labelling and symbology
- National and stand based forest inventory
- Volume estimation for forest planning purposes
- Population vs sample, normal distributions, random & stratified sampling, measurement procedures and conventions, bias and accuracy
- Measures of centre and spread, standard error, confidence intervals, t-tests, estimate precision vs desired precision in forest inventory
- Data processing and presentation

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Complete a forest stand inventory, recording forest type, productive area, species, species mix, initial spacing etc.
2. Create ArcGIS maps based on digital aerial photographs and 6" maps combined with field registrations.
3. Create and populate a subcompartment attribute table in ArcGIS.
4. Complete an inventory that quantifies the social and environmental values of a forest and the environmental risks associated with its management.
5. Apply the basic theory behind forest sampling and volume estimation for planning purposes.
6. Plan a forest inventory in terms of stratification, sampling intensity and inventory methods given certain constraints.
7. Organise and complete stand volume estimation to establish age, top height, mean dbh, stocking, basal area, mean tree volume and standing volume.
8. Prepare forest inventory reports suitable for the inclusion in a forest management plan. The reports comprise descriptive text, data analysis, tables, charts and GIS maps in an agreed format.

Learning and Teaching Methods

- Lectures: will be used to introduce the theory for forest mapping, environmental impact assessment, sampling and statistical analysis.
- Field trips: will be used to demonstrate field work procedures and for students to carry out field registration for a period of time.
- GIS practicals: will be used to generate the geodatabase for the projects and to prepare the maps involved.
- Technology: students will use statistical software to generate mean, standard deviation, standard error and confidence intervals of stands.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Project	30%	1,2,3
Project	30%	4,8
In-Class Assessment	10%	5,6
Project	30%	6,7,8,

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/ skill and understanding to carry out assessments. Failure to meet the objectives of assessments. Unable to carry out or submit independent study and research. Unable to present written work with attention to correct formatting, grammar and spelling.

40%-49%: Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on assessments. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped. Can present written work correctly formatted with minor grammar and spelling errors only.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out assessments. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design and current knowledge limitations when carrying out assessments. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on assessments. Show initiative, individual thought, and enthusiasm in self-study and independent learning.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	12	
Field Trips	48	
Lab	36	
Independent Learning	174	

Essential Material

- Mathews, R.W. and E.D. Mackey. _Forest Mensuration. A Handbook for Practitioners_. Edinburgh, UK: Forestry Commission, 2006.
- Purser, P. _Timber Measurement Manual: Standard Procedures for the Measurement of Round Timber for Sale Purposes in Ireland_. Dublin, Ireland: COFORD, 2000.

Supplementary Material

- "Forest Biodiversity Guidelines." 2000. http://www.agriculture.gov.ie/media/migration/forestry/publications/biodiversity.pdf
- "Forest Harvesting and the Environment Guidelines." 2000. http://www.agriculture.gov.ie/media/migration/forestry/publications/harvesting.pdf
- "Forest Protection Guidelines." 2002. http://www.agriculture.gov.ie/media/migration/forestry/publications/fsFPG.pdf
- "Forestry and Archaeology Guidelines." 2000. http://www.agriculture.gov.ie/media/migration/forestry/publications/archaeology.pdf
- "Forestry and the Landscape Guidelines." 2000. http://www.agriculture.gov.ie/media/migration/forestry/publications/landscape.pdf
- Forest Service, A. _Forestry Standards Manual_. Johnstown Castle Estate, co. Wexford: Department of Agriculture, Food & the Marine, 2015.
- Forest Service, A. _Forestry and Water Quality Guidelines_. Johnstown Castle Estate, co. Wexford: Department of the Marine and Natural Resources, 2000.
- Philip, M.S. _Measuring Trees and Forests_. 2nd ed.. Wallingford: CAB International, 1994.

Requested Resources

- Lecture Room: Loose Seated
- Room Type: Computer Lab